**6.8 Quiz: Polynomials Challenge — Markscheme**

1. [Maximum mark: 7]

   Consider the function $f(x) = -2(x+2)(x-4)$, for $x \in \mathbb{R}$.

   **(a)(i)** Write down the $x$-intercepts. [2]

   $x = -2$ and $x = 4$ **A1A1**

   **(a)(ii)** Find the coordinates of the vertex. [3]

   Axis of symmetry: $x = \dfrac{-2+4}{2} = 1$ **M1**

   *M1 may also be awarded for any attempt at completing the square.*

   $f(1) = -2(1+2)(1-4) = -2(3)(-3) = 18$

   Vertex: $(1, 18)$ **A1A1**

   **(b)** $f(x) = -2(x-h)^2 + k$. Write down $h$ and $k$. [2]

   $h = 1$ **A1**

   $k = 18$ **A1**

   *FT: award A1A1 for $h$ and $k$ read from the candidate's vertex in (a)(ii).*

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2. [Maximum mark: 7]

   The line $y = mx$ is tangent to the parabola $y = x^2 + 9$.

   **(a)** Show that at any intersection, $x^2 - mx + 9 = 0$. [1]

   $x^2 + 9 = mx$ **M1**

   $x^2 - mx + 9 = 0$ **AG**

   **(b)** Find the discriminant in terms of $m$. [1]

   $\Delta = (-m)^2 - 4(1)(9) = m^2 - 36$ **A1**

   **(c)** Find the values of $m$ for tangency. [2]

   Tangent: $\Delta = 0$ **M1**

   $m^2 - 36 = 0$

   $m = \pm 6$ **A1**

   **(d)** Find the coordinates of each point of tangency. [3]

For $m = 6$: $x^2 - 6x + 9 = 0$

$(x-3)^2 = 0$ **M1**

$x = 3$

$y = 6(3) = 18$. Point of tangency: $(3, 18)$ **A1**

For $m = -6$: $x^2 + 6x + 9 = 0$

$(x+3)^2 = 0$

$x = -3$

$y = -6(-3) = 18$. Point of tangency: $(-3, 18)$ **A1**

*FT: award A1 for each point correctly computed from the candidate's $x$-value.*

3. [Maximum mark: 7]

Consider the function $f(x) = 2x^2 - 8x + 3$.

**(a)** Write $f(x)$ in the form $a(x-h)^2 + k$. [3]

$f(x) = 2(x^2 - 4x) + 3$ **M1**

*M1 for any attempt at completing the square, including use of $-b/(2a)$ anywhere.*

$= 2(x^2 - 4x + 4 - 4) + 3 = 2(x-2)^2 - 8 + 3$

$= 2(x-2)^2 - 5$ **A1A1**

**(b)** Write down the coordinates of the vertex. [1]

Vertex: $(2, -5)$ **A1**

**(c)** Find the discriminant of $f(x) = 0$. [1]

$\Delta = (-8)^2 - 4(2)(3) = 64 - 24 = 40$ **A1**

**(d)** State the number of $x$-intercepts and justify. [2]

Two $x$-intercepts. **A1**

*Accept "two", "2", "double", or equivalent.*

Justification: $\Delta = 40 > 0$, so two distinct real roots. **R1**

*R1 may also be awarded for naming the two roots, or for stating that the candidate's $\Delta > 0$.*

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4. [Maximum mark: 8]

Let $p(x) = 2x^3 + x^2 - 13x + 6$.

**(a)** Show that $(x+3)$ is a factor of $p(x)$. [2]

$p(-3) = 2(-27) + (-3)^2 - 13(-3) + 6$ **M1**

$= -54 + 9 + 39 + 6 = 0$

Since $p(-3) = 0$, $(x+3)$ is a factor by the factor theorem. **A1 AG**

**(b)** Express $p(x) = (x+3)(ax^2 + bx + c)$. [2]

By polynomial division: **M1**

$2x^3 + x^2 - 13x + 6 = (x+3)(2x^2 - 5x + 2)$ **A1**

*Accept $a = 2$, $b = -5$, $c = 2$.*

*M1 may also be awarded for synthetic division (Ruffini's rule).*

**(c)** Factorise $p(x)$ completely. [2]

$2x^2 - 5x + 2 = (2x-1)(x-2)$ **M1**

*M1 may also be awarded for use of the quadratic formula.*

$p(x) = (x+3)(2x-1)(x-2)$ **A1**

*Accept* $2(x - \frac{1}{2})$ *in place of* $(2x-1)$*; do not accept* $(x - \frac{1}{2})$ *alone (missing leading coefficient).*

**(d)** Write down all solutions to $p(x) = 0$. [1]

$x = -3, \quad x = \dfrac{1}{2}, \quad x = 2$ **A1**

**(e)** Write down the $y$-intercept. [1]

$p(0) = 6$, so the $y$-intercept is $(0, 6)$. **A1**